Household Service Application

Objective:
This project is an app-based service management system designed to streamline interactions between homeowners, professionals, and admin. The system provides dashboards, search functionalities, and summaries to manage services, monitor progress, and ensure efficient communication.

Core Features:

1. **Admin Module:**
    - **Admin Dashboard (/admin_dashboard)**:
        - Manage services (create, edit, delete).
        - Approve/reject professional profiles.
        - View all service requests and statuses.
        - Access professionals' uploaded documents (PDFs).
    - **Admin Search (/admin_search)**:
        - Search and filter homeowners and professionals by attributes like ID, username, address, pincode, email, or service details.
        - Dynamic dropdowns with options fetched from the database.
    - **Admin Summary (/admin_summary)**:
        - Displays average service ratings.
        - Visual summary of service requests by status using charts.
2. **Professional Module:**
    - **Professional Dashboard (/professional_dashboard)**:
        - View service requests, ongoing services, and closed histories.
        - Accept or reject new service requests.
        - Update professional profiles.
    - **Professional Summary (/professional_summary)**:
        - Displays the professional's average rating.
        - Status-wise breakdown of service requests (e.g., Accepted, Rejected, Pending, Completed).
3. **Homeowner Module:**
    - **Homeowner Dashboard (/homeowner_dashboard)**:
        - View available services and service history.

        - Mark services as completed and provide ratings/reviews.
        - Update service requests and view professional data.
    - **Service Details (/service_details/<string:service_name>)**:
        - View details of specific services.
        - Rate professionals and rebook services.
    - **Homeowner Search (/homeowner_search)**:
        - Search for services and professionals by name, ID, rating, or location.
    - **Homeowner Summary (/homeowner_summary)**:
        - Visual summary of the logged-in homeowner's service request statuses.

Technology Stack:

1. **Frontend:**
    - **HTML/CSS +Bootstrap** for basic structure and styling.
    - **Jinja2** templating for rendering dynamic content.
2. **Backend:**
    - **Flask** (Python-based microframework) for routing, business logic, and application flow.
    - **SQLAlchemy ORM** for database interactions.
3. **Database:**
    - **SQLite** as the backend database for persisting user, service, and request data.
4. **Deployment:**
    - Local server via Flask development mode.
    - **Debug Mode Enabled** for testing and iterative development.

Student Details

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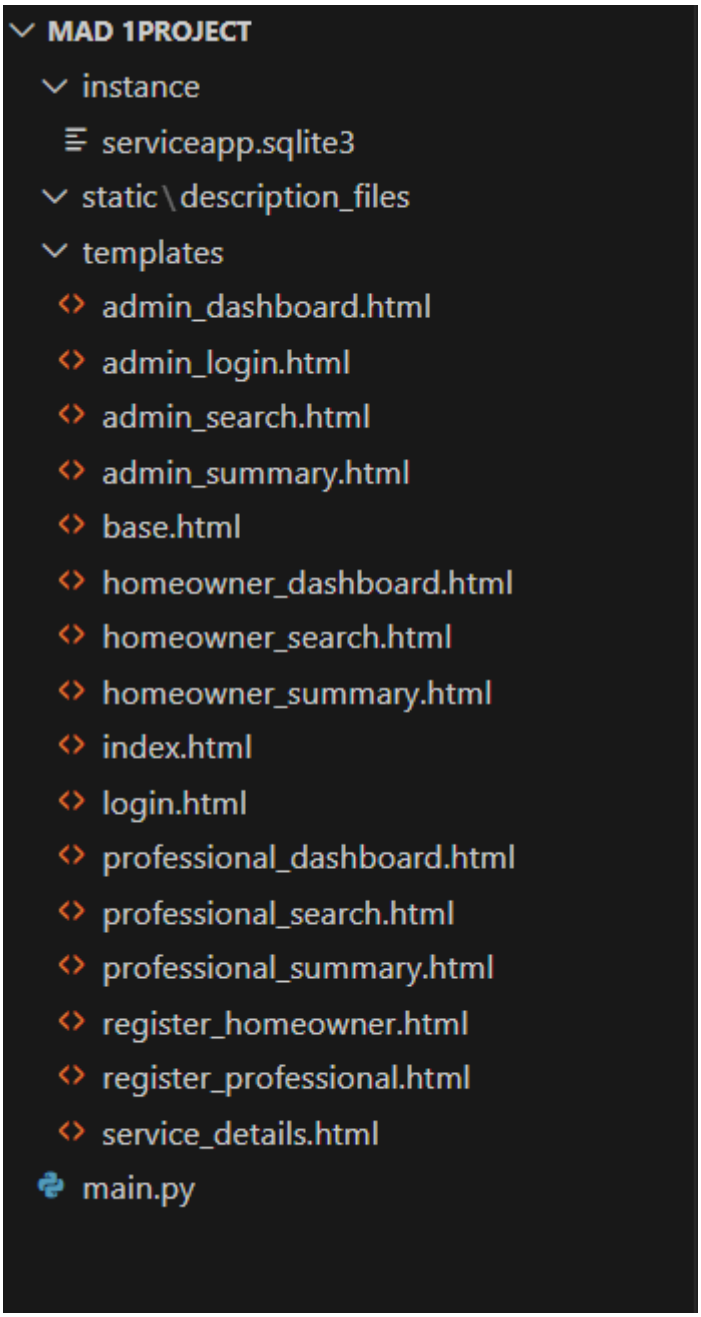
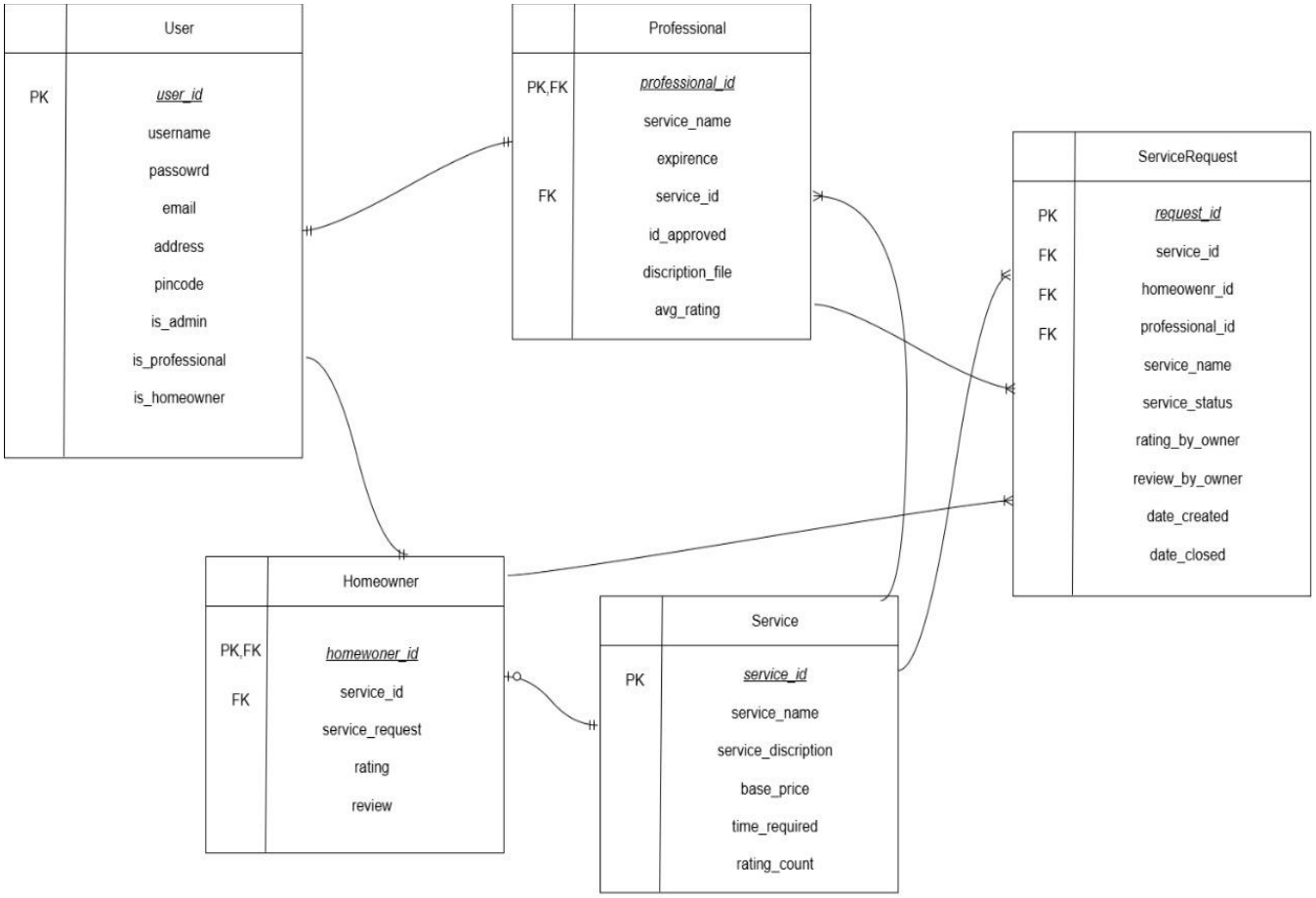
I am passionate about coding and enjoys exploring various aspects of software development. With a keen interest in problem-solving, algorithm design, and system architecture, I continuously seeks to enhance their programming skills. They are particularly intrigued by web development, backend systems, and database management, finding joy in building scalable and efficient applications. My

dedication to coding reflects a commitment to learning new technologies, improving existing solutions, and staying updated with the latest advancements in the tech world.



ER Diagram and Schema